



Department of Electrical, Electronic and Computer Engineering
BET Hons: Electrical Engineering

**Development of a Web-Based Community Energy Monitoring Dashboard with
Comparative Analytics Using Smart Circuit Controller (CBI Astute Device) Data**

(PROPOSAL)

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DECLARATION

I, Bongani Nyoni, declare that the contents of this proposal represent my own work, and that the proposal has not been submitted for academic examination towards any qualification. Furthermore, it represents my own opinions and not necessarily those of the Cape Peninsula University of Technology.

Signed

Date

Abstract

Electricity consumption in South African households and small communities is increasingly measurable, yet many readings remain difficult to interpret when they are viewed as isolated values, raw exports or device-level records. A household may see power, energy or a cost estimate, but these values do not automatically explain whether the household is improving, where peak demand is concentrated, which monitored load is contributing most to consumption, or how one site compares with an anonymised group profile. This research develops a web-based community energy monitoring dashboard that uses simulated smart circuit controller data to test comparative analytics under controlled conditions. The simulated dataset will simulate readings from smart circuit controllers with CBI-like monitoring and control capabilities, including timestamp, voltage, current, active power, interval energy, switching state, device category and data-quality flags. The project does not depend on direct access to a proprietary commercial API. Instead, CBI Astute-type devices are treated as a reference capability class for the kind of circuit-level monitoring, scheduling, load control and automation that the dataset must represent. The engineering contribution is the design and validation of a data pipeline, database structure, analytics layer, and browser-based dashboard that can later accept real smart-controller exports or, when available, approved API data. The dashboard is intended to display daily and monthly consumption, estimated cost, peak periods, monitored-load ranking, historical baseline comparison, abnormal-use warnings and anonymised peer comparison. The work is grounded in energy feedback research, which shows that electricity use must be made visible and understandable before users can manage it effectively, and it also draws on smart metering, household load-profile modelling, dashboard design, non-intrusive load monitoring and privacy-by-design literature. The prototype will be evaluated against technical and user-centred criteria, including calculation accuracy, update consistency, data completeness, missing-data handling, dashboard usability, clarity of comparative information and privacy-aware presentation. The South African relevance of the work is linked to electricity cost, household demand awareness, peak-load behaviour and the national movement towards smarter electricity measurement. The project avoids certified billing, utility switching and direct grid control. Privacy risk is reduced because the first prototype uses simulated data, while the architecture remains aligned with POPIA principles for any future real household data. Forecasting and explainable artificial intelligence are treated only as possible extensions for future versions, particularly where abnormal-use warnings or recommendations need to be explained more transparently. The main output will be a working prototype, a validated comparative analytics method, a technical evaluation report and recommendations for future real-data integration.

Keywords: CBI-like monitoring capability, smart energy monitoring, community energy dashboard, comparative analytics, smart metering, IoT, demand-side management, energy feedback, POPIA, web-based dashboard.

Mathematical Notations

- V** - RMS voltage measured by the smart device, in volts.
- I** - RMS current measured by the smart device, in amperes.
- P** - Real power consumed by a monitored load, in watts or kilowatts.
- E** - Electrical energy consumed over a selected period, in kilowatt-hours.
- Δt** - Sampling or aggregation interval used for collecting or grouping readings.
- C** - Estimated cost of consumption for a selected electricity tariff.
- B** - Historical baseline consumption for a device, household or group.
- G** - Anonymised community or peer-group average used for comparison.
- R** - Relative consumption ratio between a user's consumption and the selected benchmark.
- P_{max}** - Peak demand recorded over a selected interval.
- η** - Data completeness or valid-reading ratio, used to indicate the proportion of reliable readings available for analysis.

Definitions

Glossary of items

Algorithm - A step-by-step procedure for solving a problem using a computer

API (Application Programming Interface) - A set of rules and protocols for building and interacting with software applications, allowing different systems to communicate.

CBi Astute - A specific brand and model of smart energy monitoring device capable of measuring electrical parameters at the household level.

Dashboard - A visual interface displaying key performance indicators and data insights, often using charts, graphs, and gauges.

Database Schema - The structural definition of a database, including tables, fields, relationships, and constraints.

Community energy monitoring-The combined monitoring of several households, rooms, buildings or selected loads so that collective consumption patterns can be understood.

Comparative Analytics - The process of comparing data sets across different entities (e.g., households) to identify patterns, outliers, and benchmarks.

Demand-side management - Actions that help users reduce, shift or better understand electricity demand instead of only increasing supply.

Personalised Recommendation - Tailored advice generated for a specific user based on their unique consumption patterns and historical data.

Nomenclature

ADA – Advanced Distribution Automation

API - Application Programming Interface

CBi - CBi Electric (manufacturer of the Astute device)

API - Application Programming Interface

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CPU - Central Processing Unit

CSS - Cascading Style Sheets

GUI - Graphical User Interface

HTML - HyperText Markup Language

HTTP - HyperText Transfer Protocol

IoT - Internet of Things

JSON - JavaScript Object Notation

kWh - Kilowatt-hour

NoSQL - Not Only Structured Query Language

REST - Representational State Transfer

SQL - Structured Query Language

UI - User Interface

UX - User Experience - Central Processing Unit

CSS - Cascading Style Sheets

GUI - Graphical User Interface

HTML - HyperText Markup Language

HTTP - HyperText Transfer Protocol

IoT - Internet of Things

JSON - JavaScript Object Notation

kWh - Kilowatt-hour

NoSQL - Not Only Structured Query Language

REST - Representational State Transfer

SQL - Structured Query Language

UI - User Interface

UX - User Experience

Table of Contents

Abstract.....	ii
Mathematical Notations.....	iii
Definitions.....	iv
Nomenclature.....	v
List of Figures	10
List of Tables.....	10
Section A: Introduction	11
1. Introduction.....	11
2. Awareness of problem.....	12
3. Problem statement.....	13
2. Research aim and objectives	13
2.1 Aim.....	13
2.2 Research objectives.....	13
2.3 Research questions	14
3. Hypothesis.....	14
4. Motivation of the research project	14
5. Assumptions.....	15
6. Chapter breakdown.....	15
Section B: Literature review.....	15
1. South African residential energy context.....	15
2. Smart metering, CBI Astute data and IoT data acquisition	16
3. Energy feedback and comparative analytics	18
4. Web-based dashboard design and data processing.....	19
5. Smart-controller dataset simulation and controlled validation	20
6. Privacy, security and ethical handling of smart energy data.....	22
7. Literature review conclusion.....	22
Section C: Design and Implementation	24
1. Monitoring and analytics scheme	24
2. System requirements	24
3. Simulated smart-controller data specification and generation plan	25
4. Data model and algorithms	26
5. Dashboard interface	28
6. Implementation tools	29
7. Testing and evaluation.....	29

8. Ethical and regulatory considerations.....	31
9. Risks and mitigation	32
10. Expected deliverables.....	32
11. Scope and limitations.....	33
12. Work plan.....	33
13. Conclusion	34
BIBLIOGRAPHY.....	35

List of Figures

Figure 1: Research Logic For Community Energy Dashboard	12
Figure 2: Simulated smart circuit controller	17
Figure 3: End-to-end system architecture.	20
Figure 4: Privacy-by-design layer for smart energy data.	22
Figure 5: Core relational data model.	26
Figure 6: Comparative analytics pipeline.....	28
Figure 7: Dashboard wireframe for prototype evaluation.....	29
Figure 8: Evaluation logic for the prototype.....	31

List of Tables

Table 1: Technical comparison of data-source options.....	17
Table 2: Literature comparison on feedback, datasets and analytics	18
Table 3: Simulated smart-controller dataset field specification	21
Table 4: Functional and non-functional requirements.	24
Table 5: Core database entities.	27
Table 6: Testing and evaluation matrix.....	30
Table 7: Risks and mitigation.	32
Table 8: Project work plan from February to October 2026.....	33

Section A: Introduction

1. Introduction

Electricity use in a household is often treated as a monthly bill rather than a measurable operating pattern. That creates a gap between the amount of energy data that can be collected and the amount of insight that ordinary users can act on. In South Africa, this gap matters because households rely on electricity for cooking, refrigeration, heating, lighting, communication devices, security systems and small business activity, while electricity cost and peak demand remain practical constraints. Statistics South Africa reported that access to mains electricity increased from 76.7% in 2002 to 90.2% in 2024, which confirms that electricity dependence is now widespread enough for household-level monitoring to have practical value (Statistics South Africa, 2025).

Smart meters and smart circuit controllers make household electricity use more visible, but visibility on its own is not enough. A reading of power or energy does not tell the user whether the household is improving, whether a load is dominating consumption, whether a repeated evening peak is forming, or whether a site is above the average profile of a comparable group. Smart circuit controllers with CBI-like capability provide the practical reference point because they combine energy monitoring, scheduling, switching control, load management and automation using variables such as current, voltage, time and power (CBI-electric, 2026a). The useful engineering layer sits between those readings and user interpretation: data validation, storage, aggregation, comparison, visualisation and privacy control.

This project develops a web-based community energy monitoring dashboard using simulated smart circuit controller data. The shift to simulation removes dependency on a specific commercial device or API while keeping the data close to the operating logic of CBI Astute-type controllers. The simulated dataset will mimic realistic timestamped readings such as voltage, current, active power, energy consumption, device category, switching state and data-quality flags. This keeps the prototype feasible while preserving the same dashboard logic that would be needed for future real-device integration.

The dashboard will receive structured smart-controller-style data, validate it, store it in a relational database, apply comparative analytics and present the results through a browser-based interface. The project does not replace certified meters, municipal billing, Eskom smart-meter infrastructure or electrical load-control systems. It creates a research-ready interpretation layer that turns circuit-level energy readings into baseline comparison, peak-load visibility, device ranking, estimated cost and anonymised peer comparison.



Figure 1: Research Logic For Community Energy Dashboard

2. Awareness of problem

Household energy data is useful only when the user can understand what the data means. A raw table may contain valid voltage, current, power and energy values, but it does not automatically show whether the home is using more energy than normal, when unnecessary demand occurs, which monitored load is responsible, or whether the household is consuming more than a comparable group. Darby (2006) argues that domestic energy use is often invisible to users and that feedback is needed to make consumption understandable and controllable. That argument remains central to this project.

South Africa adds a specific operating context. Smart metering is moving into the formal electricity system, with Eskom stating that its smart meter programme started in 2025 and is expected to continue until 2029 (Eskom, 2026). Even where formal utility metering improves, a household or small community still needs local interfaces that explain consumption patterns, highlight peaks and compare behaviour in practical terms. The dashboard therefore sits below utility-grade infrastructure and focuses on awareness, analytics and interpretation rather than revenue-grade measurement.

The data-access problem also matters. A project that depends on a specific commercial API can become weak if the device, credentials or developer route are not available. Using simulated or mimicked smart-controller data gives the research a stable foundation. It allows known scenarios to be created, such as missing readings, peak-use events, abnormal geyser behaviour, high standby consumption, switching-state changes and community-level variation. These scenarios can be used to test the dashboard properly before future real-data integration is attempted.

Energy data must also be treated carefully because it can reveal household routines. Smart meter privacy literature shows that electricity readings can expose occupancy patterns and behavioural details (Asghar et al., 2017; Beckel et al., 2014). The first prototype avoids direct household exposure by using simulated data, but the architecture still needs privacy-by-design because future versions may process real user data

3. Problem statement

Household electricity readings are often difficult to interpret when they are viewed only as isolated values, raw exports or spreadsheet-style records. A user may see voltage, current, active power or accumulated energy, yet these readings do not automatically show whether consumption is improving, which monitored load contributes most to usage, when peak demand occurs, or how one household compares with an anonymised group profile.

The research problem is the lack of a privacy-aware web dashboard that can import structured smart-controller energy readings, validate and store them, apply comparative analytics and present meaningful individual and community-level energy insights. Since direct access to a specific commercial smart-device API is not available, the prototype will use a controlled simulated smart circuit controller dataset that mimics realistic circuit-level energy measurements. The core problem remains the absence of a clear interpretation layer that turns energy readings into baseline, peak-load, device-ranking, estimated-cost and anonymised peer-comparison information.

2. Research aim and objectives

2.1 Aim

The aim of the research is to design, develop and evaluate a web-based community energy monitoring dashboard that uses simulated smart circuit controller data to provide comparative analytics for household or small-community electricity consumption. The system will focus on data ingestion, validation, storage, baseline comparison, peak detection, device ranking, estimated cost and anonymised peer comparison. Forecasting and explainable alert logic may be considered only as optional extensions where the simulated dataset is strong enough to support them.

2.2 Research objectives

This study is guided by the following five objectives:

1. **Data specification:** Identify and define the electrical variables and dataset structure required to support a smart circuit controller-based community energy monitoring dashboard. A formal data specification document defining at minimum eight variables voltage, current, active power, cumulative energy, UTC timestamp, controller channel ID, device category, and switching state including acceptable value ranges, units, data-quality thresholds, and a schema diagram.
2. **Data pipeline:** Design and implement a secure data pipeline that ingests, validates, stores, and pre-processes smart circuit controller energy readings for dashboard consumption. A functional ingestion and storage layer including schema-enforced database tables, validation rule definitions, missing-value handling procedures, and data-quality flags demonstrated by processing the simulated dataset with zero unhandled errors and a documented audit log.
3. **Analytics development:** Develop comparative analytics algorithms to derive interpretable energy metrics, including consumption baselines, peer-group comparisons, peak-demand detection, device-level rankings, anomaly warnings, and estimated cost calculations. A documented analytics module with at least six implemented algorithms, each accompanied by its formula, input parameters, output format, and a unit-tested verification result against a known reference case, with accuracy validated to within $\pm 5\%$ of manual calculations.
4. **Dashboard prototype:** Build a responsive, web-based dashboard prototype that presents individual and community-level energy information through interactive visualisations and

exportable summaries. A browser-deployable prototype incorporating at minimum five chart types, four KPI cards, a peer-comparison panel, and a CSV/PDF export function verified to render correctly on desktop and mobile viewports (≥ 320 px width) without layout breakage.

5. **Evaluation:** Evaluate the dashboard prototype against criteria of calculation accuracy, interface usability, responsiveness, data completeness, and privacy-aware data presentation. A written evaluation report documenting: accuracy test results for each analytics module; usability feedback from at least five participants using a standardised instrument such as the SUS; cross-device responsiveness test outcomes, a data-completeness audit; and identified limitations with prioritised improvement recommendations.

2.3 Research questions

The project will be guided by the following research questions:

- Which electrical variables and metadata fields are required to represent smart circuit controller energy data in a dashboard-ready format?
- How can simulated smart circuit controller readings be generated so that they are realistic enough for testing ingestion, validation, analytics and visualisation?
- How can raw timestamped readings be structured into a reliable data model for household and community analytics?
- Which comparative metrics are most useful for identifying high usage, peak periods, abnormal consumption and monitored-load ranking?
- How can web dashboard present energy feedback clearly enough for users to interpret consumption patterns without reading raw data tables?
- What privacy and security controls are required if the architecture is later connected to real smart-controller or household energy data?

3. Hypothesis

H1: A web-based dashboard that uses structured simulated smart circuit controller data and comparative analytics will improve the interpretability of household and community energy consumption patterns compared with viewing isolated raw readings or spreadsheet-style data.

H0: A web-based dashboard using structured simulated smart circuit controller data and comparative analytics will not materially improve the interpretability of household and community energy consumption patterns compared with viewing isolated raw readings or spreadsheet-style data.

4. Motivation of the research project

The motivation is both technical and practical. Technically, the project creates a complete engineering chain from smart-controller-style energy data representation to dashboard interpretation. It requires knowledge of electrical quantities, time-series data, databases, web systems, analytics, privacy and prototype evaluation. Practically, the project responds to the need for South African households and small communities to understand consumption behaviour more clearly, especially where energy cost and peak usage affect daily decisions.

A simulated smart-controller dataset is not a downgrade when the research is framed correctly. It gives the prototype-controlled input conditions, which makes it possible to test whether the dashboard handles expected readings, abnormal readings, missing intervals, switching-state changes and peer-comparison cases. Real data can be introduced later if

approved access becomes available, but the first validation does not need to wait for a commercial API.

The project is realistic for an Honours-level engineering study because it does not depend on building a certified meter or gaining access to a private cloud ecosystem. It focuses on the part that can be designed, implemented and evaluated within the project period: data structure, ingestion, validation, storage, comparative analytics, dashboard interface and privacy-aware reporting. The same architecture can later support campus monitoring, student accommodation monitoring, municipal awareness programmes or community energy education.

5. Assumptions

- A controlled simulated smart circuit controller dataset will be developed for prototype testing.
- The simulated dataset will contain realistic variables such as timestamp, voltage, current, active power, interval energy, cumulative energy, device category, switching state and data-quality flag.
- The first prototype will not require direct access to a commercial smart-device API.
- The architecture will remain data-source agnostic so that future versions can connect to real smart-controller data, approved exports or APIs if access becomes available.
- The dashboard will not be used for certified billing, tariff enforcement, utility switching or direct grid control.
- Estimated cost calculations will use configurable tariffs and will not claim to replace official municipal or Eskom billing.
- Community comparison will use anonymised or pseudonymised identifiers and will not expose household names, addresses or detailed occupancy routines.
- The technical evaluation will prioritise calculation correctness, reliability, usability, responsiveness, data completeness and privacy-aware presentation.
- Forecasting, SHAP-based explanation or machine-learning recommendations will remain optional and will not be required for the core comparative dashboard.

6. Chapter breakdown

Section A establishes the research context, problem, aim, objectives, hypothesis and assumptions. Section B reviews South African energy conditions, smart metering, smart circuit controller capability, household energy datasets, energy feedback, comparative analytics, dashboard systems and privacy. Section C presents the design and implementation plan, including the architecture, simulated smart-controller dataset specification, data model, algorithms, dashboard interface, evaluation method, risks, work plan and bibliography. Forecasting and explainability appear only as possible extensions to the analytics layer where they support abnormal-use explanation or future recommendation work.

Section B: Literature review

1. South African residential energy context

The South African household energy context supports the need for practical monitoring tools. Increased access to electricity has expanded the number of households using electrical appliances, but greater dependence also makes consumption behaviour more important. The

2024 General Household Survey shows that 90.2% of households were connected to mains electricity in 2024, compared with 76.7% in 2002 (Statistics South Africa, 2025). The consequence is not only improved access; it is also a stronger need for visibility at household and community level.

Smart metering is becoming part of the formal electricity system. Eskom states that its smart meter programme started in 2025 and will continue until 2029 (Eskom, 2026). This confirms that smarter measurement is part of the current national direction. However, a utility rollout does not automatically solve local interpretation. Users still need interfaces that explain consumption patterns, highlight peak periods and compare behaviour in a way that is meaningful at household scale.

South African load-profile research also supports the value of comparison. Toussaint (2020) describes the Domestic Electrical Load Metering Hourly dataset as a major South African household electricity database containing millions of daily load profiles over a twenty-year period. The use of clustering and representative daily load profiles shows that consumption behaviour can be grouped into archetypes, which supports the idea that comparative analytics can move beyond raw readings.

Ritchie, Engelbrecht and Booysen (2022) further demonstrate the value of data-driven household electricity profiles by using smart meter data from South African households to create statistically representative demand profiles. Their work does not solve the dashboard problem directly, but it supports the logic that household energy data can be structured, compared and interpreted when the correct analytical layer is used.

2. Smart metering, CBI Astute data and IoT data acquisition

Smart metering and consumer IoT devices both collect electrical measurements, but they serve different roles. Utility smart meters are normally part of formal AMI and billing infrastructure, while consumer smart devices support local monitoring, control and automation. The dashboard in this research sits closer to the consumer-device side. It is not intended to certify billing; it uses accessible device data to support energy awareness and comparison.

The CBI Astute Smart Controller is relevant because it combines monitoring, scheduling, load management and automation. CBI states that the device is Wi-Fi enabled, supports scheduled energy management, provides energy monitoring and can respond to variables such as current, voltage, time and power (CBI-electric, 2026a; CBI-electric, 2026b). These variables are sufficient for a dashboard prototype because power and energy can support trend analysis, cost estimation, peak detection and baseline comparison.

International IoT smart-meter work supports the selected variable set. Munoz et al. (2022) developed and validated an IoT smart meter with load control that measured RMS voltage, RMS current, active, reactive and apparent power, energy and power factor, with experimental errors below 1% across the tested electrical parameters. Their work is not used as a hardware template for this project, because CBI Astute devices already provide the consumer measurement source, but it supports the engineering logic that a useful home-energy platform should process more than kWh alone and should validate measurement quality before using the data for feedback.

Data acquisition must be handled carefully. The preferred route is approved access through the CBI Home ecosystem or supported export methods. Because CBI Astute devices are associated with the Tuya-based smart-home ecosystem in several support contexts, Tuya documentation is relevant for understanding possible data structures. Tuya describes energy metering as a feature that collects operating power, voltage, current and energy consumption

for reporting to the Tuya Cloud (Tuya Developer, 2024a). Tuya also documents energy consumption management APIs that can query energy use by device and by period, including day, week, month and year (Tuya Developer, 2024b).

The project must not assume unrestricted access to private cloud data. A technically sound proposal must allow for three acquisition options: direct approved API access, user-authorised export or manual import of logged readings. This keeps the design feasible even if API access is delayed or limited. It also protects the project from depending on unofficial reverse engineering, which would weaken the ethical and academic standing of the work.

Table 1: Technical comparison of data-source options.

Data-source option	Strength	Limitation	Use in this project
Simulated smart circuit controller dataset	Fully controlled, repeatable and not dependent on device/API access.	Cannot prove live-device reliability or real household behaviour.	Primary route for prototype development and validation.
Manual CSV import	Simple to implement and useful for demonstrations or future exported controller readings.	May require cleaning and does not provide real-time updates.	Supported as the first ingestion route.
Approved controller API route	Can support automated updates and device-level integration.	May require credentials, platform approval and stable permissions.	Future extension if approved access becomes available.
Public energy datasets	Useful for checking realistic load-profile shapes and appliance-level structures.	Usually not South African and may not match local tariffs or household behaviour.	Used as literature support and modelling reference, not as the main dataset.

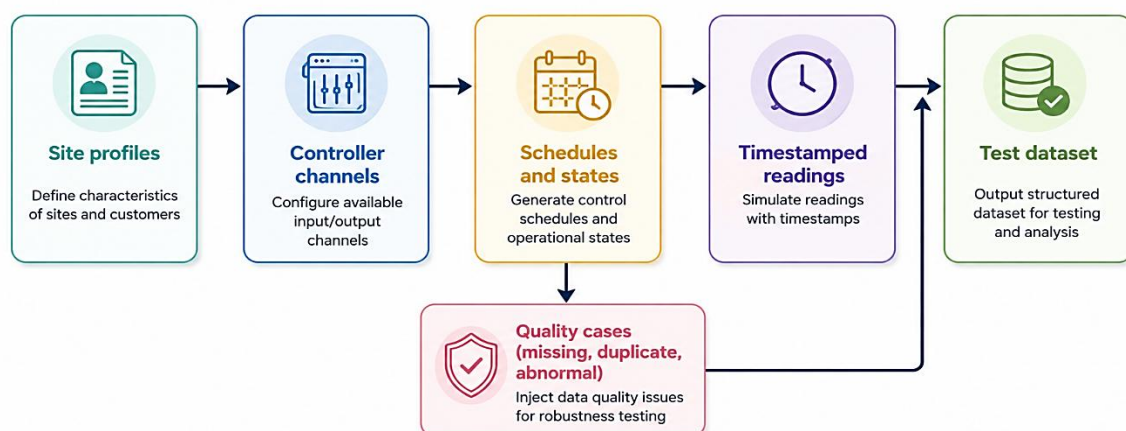


Figure 2: Simulated smart circuit controller

3. Energy feedback and comparative analytics

Energy feedback research is the strongest scholarly foundation for this topic. Darby (2006) argues that feedback makes energy visible and supports learning about how consumption can be controlled. Fischer (2008) adds that effective feedback depends on frequency, duration, content, breakdown, medium, presentation, comparisons and combination with other instruments. This directly supports a dashboard approach because the dashboard is not only a display; it is the mechanism that organises frequency, comparison, breakdown and presentation.

Meta-level evidence also supports the importance of feedback. Ehrhardt-Martinez, Donnelly and Laitner (2010) reviewed advanced metering and feedback programmes and found that feedback can create household electricity-saving opportunities when it is designed and delivered effectively. Karlin, Zinger and Ford (2015) later showed through meta-analysis that the effect of energy feedback depends on design and context. The implication is that the dashboard must be evaluated as a feedback system, not merely as a database viewer.

Comparative analytics gives the dashboard its distinctive value. Allcott (2011) found that social comparison reports reduced average household energy consumption in a large field experiment. Schultz et al. (2007) showed that social norms can influence household energy behaviour, although feedback must be designed carefully to avoid encouraging low consumers to increase usage. This project therefore uses comparison as an informative benchmark rather than as a public ranking that exposes or shames users.

The comparative analytics will include four levels: self-comparison against previous use, device comparison inside the same household, anonymised peer comparison across the community and threshold comparison against configured warning levels. This structure suits simulated data because each scenario can be controlled and checked against manual calculations before the dashboard output is accepted.

Table 2: Literature comparison on feedback, datasets and analytics

Source	Main contribution	Design implication for the dashboard
Darby (2006)	Energy is often invisible and feedback supports self-learning.	The dashboard must translate readings into understandable feedback.
Fischer (2008)	Effective feedback depends on frequency, breakdown, medium, comparison and presentation.	The prototype must include charts, breakdowns and comparison, not only raw readings.
Allcott (2011)	Neighbour comparison can reduce average energy consumption.	Peer comparison should be anonymized and carefully framed.

Source	Main contribution	Design implication for the dashboard
Schultz et al. (2007)	Social norms can influence household behaviour but can also create rebound effects.	The dashboard should avoid public shaming and protect low-consumption users from negative comparison effects.
Kelly and Knottenbelt (2015)	UK-DALE shows how household and appliance-level electricity data can be structured as time-series records.	Supports the dataset structure and the need for appliance or monitored-load labels.
Murray, Stankovic and Stankovic (2017)	REFIT shows the value of longitudinal household electrical load measurements.	Supports testing with longer time periods and repeated daily patterns.
Asghar et al. (2017)	Smart meter data can reveal private household information.	The architecture must include privacy-by-design even when real data is only a future extension.

4. Web-based dashboard design and data processing

A web dashboard is suitable because it can be accessed from different devices, supports centralised data storage and can visualise trends using familiar charts. The interface should prioritise interpretation over technical clutter. Users should see the current consumption position, the main contributors to energy use, the peak period, the cost estimate and the comparison result without needing to inspect raw data.

The data processing layer is as important as the interface. Raw readings may contain missing timestamps, repeated values, communication gaps, out-of-range values or device-state changes. The system must therefore validate timestamps, remove duplicates, flag missing intervals, aggregate readings into usable periods and preserve audit trails. For simulated smart-controller data, those quality conditions can be inserted deliberately so that the validation layer can be tested under known conditions.

Non-Intrusive Load Monitoring literature is relevant but not central to the first prototype. Hart (1992) established NILM as a method of identifying appliance loads from aggregate measurements. In this project, simulated smart-controller channels can be labelled directly, so the first version does not need advanced disaggregation. NILM remains useful as a future extension where only aggregate household data is available and appliance-level estimates are needed.

The dashboard architecture will separate data source, ingestion, validation, storage, analytics and visualisation. This separation makes the project easier to test and extend. A change from simulated smart-controller CSV data to future real-device data should not require rewriting the analytics and dashboard layers. Likewise, a new chart or metric should not require changing the ingestion source.

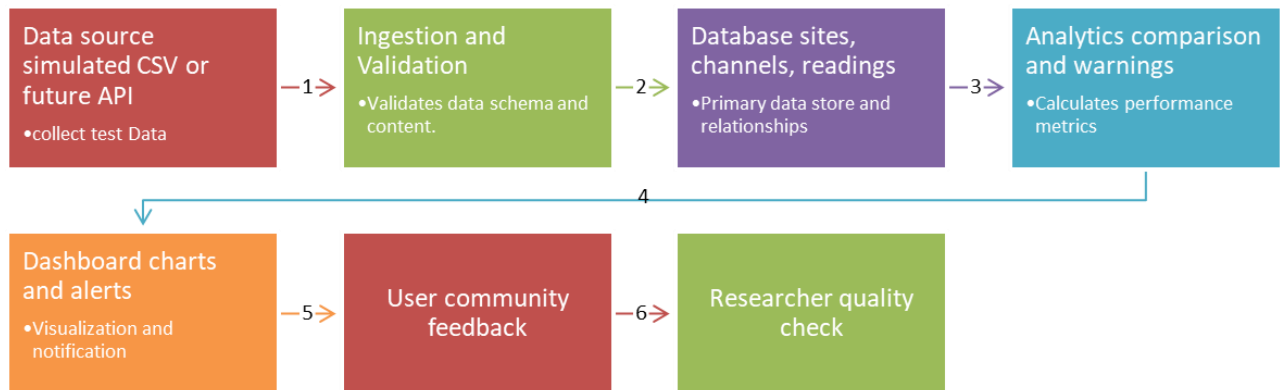


Figure 3:End-to-end system architecture.

5. Smart-controller dataset simulation and controlled validation

The simulated smart-controller dataset is central to the corrected project scope. It will not be random numbers without engineering meaning. The dataset will be generated from site profiles, controller

channels, monitored-load categories, time-of-day schedules, weekday/weekend patterns and controlled abnormal events. This allows the dashboard to be tested against realistic operating logic while keeping the data source independent from a commercial API.

The minimum dataset should contain at least ten household or site profiles, five to six monitored load categories and thirty days of timestamped readings at a 15-minute interval. This is large enough to test daily trends, monthly summaries, peak periods, missing readings, controller-channel ranking and group comparison. A larger stress-test dataset can then be generated to check dashboard responsiveness and database performance.

The electrical variables must remain physically sensible. Voltage should remain within a realistic low-voltage supply range with minor variation, current should follow the device power profile, and active power should be internally consistent with voltage, current and a selected power factor where those variables are simulated together. Energy per interval should be calculated from active power and interval length so that the dashboard can be checked against known manual totals.

Controlled validation is stronger than pretending the data is live. Known input conditions allow the researcher to confirm that the dashboard calculates totals, peaks, ratios and flags correctly. Once the prototype is stable, the same schema can accept real smart-controller exports or approved API readings if access becomes available

Table 3: Simulated smart-controller dataset field specification

Field	Meaning	Type	Use in dashboard
<i>reading_id</i>	Unique record identifier.	<i>String / integer</i>	Required for traceability.
<i>site_id</i>	Anonymized household, room, building or test site identifier.	<i>String</i>	No personal address stored.
<i>device_id</i>	Identifier for the monitored load or simulated smart circuit controller channel.	<i>String</i>	Links readings to controller channel and device category.
<i>controller_channel</i>	Channel or load category such as geyser, fridge, lighting, plugs, cooking or background load.	<i>Text</i>	Used for ranking and comparison.
<i>timestamp</i>	Date and time of reading.	<i>Date-time</i>	Used for trend, peak and interval calculations.
<i>voltage_v</i>	RMS voltage value.	<i>Numeric</i>	Range checks applied.
<i>current_a</i>	RMS current value.	<i>Numeric</i>	Derived or simulated consistently with power.
<i>power_kw</i>	Active power in kilowatts.	<i>Numeric</i>	Primary value for energy calculation.
<i>energy_kwh_interval</i>	Energy consumed during the interval.	<i>Numeric</i>	Calculated from power and interval length where needed.
<i>cumulative_energy_kwh</i>	Running energy total where simulated or imported.	<i>Numeric</i>	Optional but useful for consistency checks.
<i>switching_state</i>	Controller or device state such as on, off, standby or fault.	<i>Text</i>	Supports abnormal-use interpretation.
<i>quality_flag</i>	Valid, missing, duplicate, out-of-range or abnormal event.	<i>Text</i>	Prevents poor data from silently distorting analytics.

6. Privacy, security and ethical handling of smart energy data

Smart energy data is not neutral. It can reveal when people are at home, when appliances are used and when a household follows predictable routines. Asghar et al. (2017) show that smart meter data can expose private information, while Beckel et al. (2014) show that household characteristics can be inferred from energy data. Even though the prototype uses simulated data, the system should be designed as if future real household data may be connected.

South African law strengthens this requirement. POPIA sets out conditions for lawful processing, including accountability, processing limitation, purpose specification, further processing limitation, information quality, openness, security safeguards and data subject participation (Republic of South Africa, 2013). The dashboard design therefore needs privacy controls at the architecture level, not only in the ethics form.

For this project, privacy-by-design means that the prototype will not require real household identities. Site identifiers will be anonymised. Community views will show aggregated comparisons rather than household names. Raw data access will be separated from dashboard views. If real data is introduced later, participation must require informed consent, clear purpose specification, secure storage, role-based access and the right to withdraw where applicable.

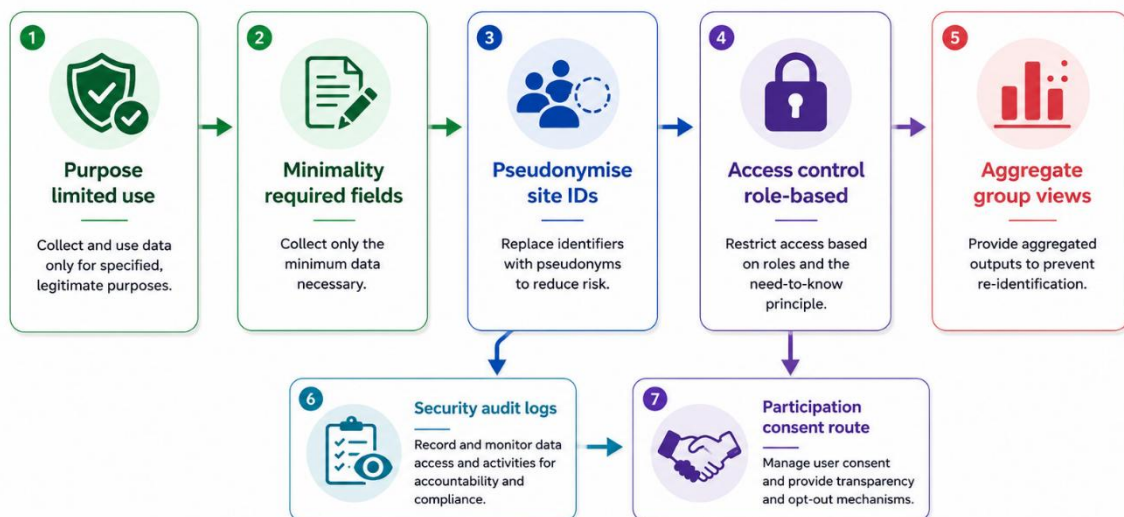


Figure 4: Privacy-by-design layer for smart energy data.

7. Literature review conclusion

The literature supports the need for a dashboard that does more than display readings. South African energy conditions create a practical need for household-level awareness, while smart-meter rollouts show that measurement is becoming more important in the national electricity environment. CBI-like smart circuit controller capability gives the project a practical device class to mimic without making the prototype dependent on that device. Feedback scholars show that users need clear, frequent and comparative information. Household energy datasets show that time-series load profiles can be structured and analysed. Privacy scholars and POPIA show that household energy data must be handled carefully. The research gap is therefore a practical engineering gap: a privacy-aware web dashboard that can turn structured smart-controller-style readings, initially simulated, into comparative community energy insight.

Section C: Design and Implementation

1. Monitoring and analytics scheme

The monitoring scheme begins with structured simulated smart-controller readings and ends with dashboard feedback. The core variables are timestamp, voltage, current, active power, interval energy, cumulative energy, device category, switching state and data-quality flag. These readings will be generated or imported in a common format and normalised into a relational data model. The analytics layer will then produce consumption summaries, baselines, comparisons, peak indicators, cost estimates and warning flags.

The dashboard will use a layered architecture. The data-source layer supplies simulated smart-controller CSV records or future approved readings. The ingestion layer receives the records. The validation layer checks time, range, duplicates and missing intervals. The database layer stores device, site, reading and aggregation records. The analytics layer calculates comparison outputs. The presentation layer displays the results through a web interface. This design keeps the prototype testable because each layer can be verified separately.

The dashboard will use a layered architecture. The device layer collects or exports readings. The ingestion layer validates and timestamps the readings. The database layer stores device, household, reading and aggregation records. The analytics layer generates comparison and warning outputs. The presentation layer displays the results through a web interface. This design keeps the prototype testable because each layer can be verified separately.

2. System requirements

Table 4:Functional and non-functional requirements.

Requirement type	Requirement	Reason
<i>Functional</i>	Import simulated smart circuit controller readings from CSV or generated datasets.	The dashboard depends on structured timestamped smart-controller-style readings.
<i>Functional</i>	Store readings by timestamp, site, device and measured variable.	Structured storage is needed for comparison and trend analysis.
<i>Functional</i>	Display daily, weekly and monthly consumption graphs.	Users need time-based visibility.
<i>Functional</i>	Calculate baseline comparison and peer-group comparison.	Comparative analytics is the main research contribution.
<i>Functional</i>	Identify peak demand periods and high-consumption devices.	Peak visibility supports practical demand-side awareness.

<i>Functional</i>	Generate estimated cost using a configurable tariff.	Cost makes energy readings more meaningful to users.
<i>Functional</i>	Flag missing, duplicate and out-of-range readings.	Data quality affects the reliability of analytics.
<i>Non-functional</i>	Protect user identity through pseudonymised identifiers and role-based access.	Household and controller-level energy data is sensitive if real data is later used.
<i>Non-functional</i>	Provide a responsive web interface for laptop and mobile viewing.	Community users may access the dashboard from different devices.
<i>Non-functional</i>	Keep the system modular so the data source can move from simulation to future smart-controller API or export data.	The project must remain useful beyond the first simulated prototype.
<i>Optional extension</i>	Provide transparent abnormal-use explanations through comparative rules or interpretable forecasting where data quality permits.	Keeps possible XAI work as support for the dashboard rather than the main research contribution.

3. Simulated smart-controller data specification and generation plan

The dataset will be generated using household or site profiles and smart-controller channel usage rules. Each site will receive a profile such as low, medium or high usage. Each controller channel or device category will receive a typical load range, operation pattern, switching state and random variation. Weekday and weekend behaviour will be separated because household energy patterns often differ between workdays and non-workdays.

The first dataset should use a 15-minute interval because it is detailed enough to show peak behaviour while remaining manageable for a web prototype. For thirty days, one site produces 2 880 time intervals. With ten sites and six monitored smart-controller channels, the dataset can produce more than 170 000 device-level rows, which is enough to test ingestion, aggregation, dashboard rendering and peer comparison. The dataset can be extended later for stress testing.

Abnormal-use scenarios will be inserted deliberately. Examples include a geyser channel running for longer than normal, a fridge showing unusually high cycling, a plug-load channel staying on overnight, missing readings caused by a simulated communication gap and duplicate readings caused by repeated imports. These cases allow the warning and data-quality logic to be tested against known expected outputs.

$$P = V \times I \times PF$$

Where active power is simulated from voltage and current, the power factor will be selected according to the device category. Where active power is generated directly, voltage and current will be calculated or checked so that the reading remains physically sensible.

$$E = \Sigma(P \times \Delta t)$$

For interval energy, P is expressed in kilowatts and Δt in hours. For a 15-minute interval, $\Delta t = 0.25$ hours. This relationship allows the dashboard totals to be checked manually.

$$C = E \times \text{tariff}$$

The cost value is an estimate only. It will use a configurable tariff and will not be presented as an official bill.

4. Data model and algorithms

The database will use a relational structure because the data has clear entities: users, sites, smart-controller channels, readings, daily summaries, comparison results and audit logs. A lightweight prototype can use SQLite during early development and migrate to PostgreSQL if multi-user testing or larger datasets require stronger concurrency. The data model will not store unnecessary personal details. Sites can be represented by research identifiers and community views can remain aggregated.

The analytics layer will convert readings into metrics that the dashboard can display. Daily totals will be calculated from interval energy. Peak power will be identified per device, site and period. Device ranking will sort monitored loads by energy consumption. Baseline comparison will compare current consumption with previous similar periods. Peer comparison will compare a household against an anonymised group average. Data completeness will show whether the result is reliable enough to interpret.

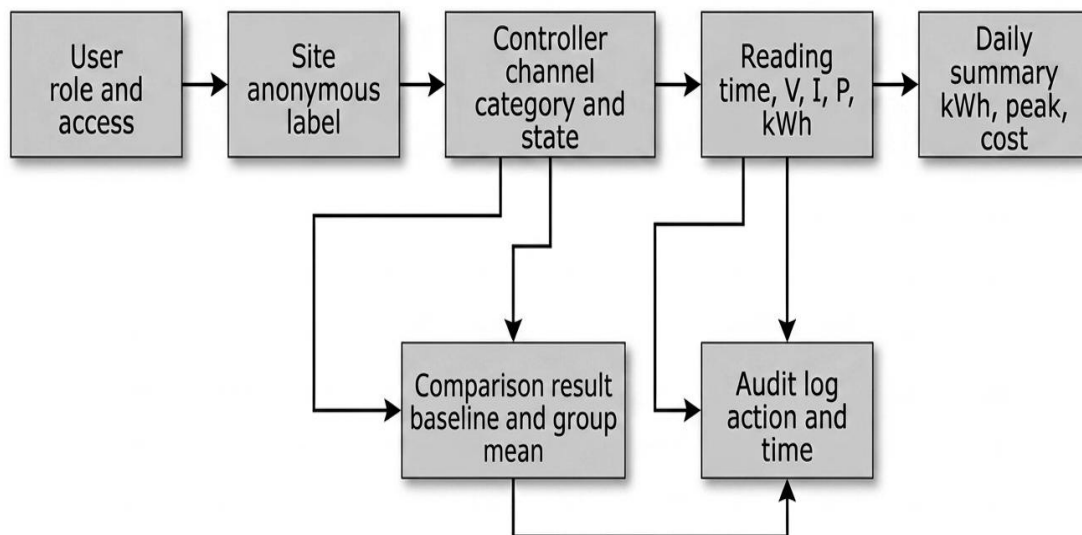


Figure 5:Core relational data model.

Table 5:Core database entities.

Entity	Key fields	Purpose
User	user_id, role, consent_status, access_level	Controls authorized access and consent status.
Site	site_id, community_id, anonymised_label, site_type	Groups readings without exposing a personal address or identity.
Smart controller channel	device_id, site_id, controller_channel, device_category, rated_power, switching_state	Links readings to the simulated or future monitored controller channel.
Reading	reading_id, device_id, timestamp, voltage, current, power, energy_kWh, state, quality_flag	Stores raw or cleaned measurement values.
Daily summary	summary_id, device_id, date, total_kWh, peak_power, cost_estimate	Supports dashboard trend and cost views.
Comparison result	comparison_id, site_id, period, baseline_kWh, group_average_kWh, ratio, status_flag	Stores comparative analytics outputs.
Audit log	event_id, user_id, action, timestamp	Tracks access and changes for accountability.

- Baseline: B is calculated as the mean consumption for the same site, device category and comparable historical period.
- Peer average: G is calculated from anonymized sites in the same group and period, excluding the current site where required.
- Relative ratio: $R = E_{user} / G$. A value above 1 indicates consumption above the group average.
- Peak flag: a reading or interval is flagged where P exceeds a configured threshold or falls within the highest percentile of the user profile.
- Device ranking: monitored devices are sorted by total kWh over the selected period.
- Data completeness: $\eta = \text{valid readings received} / \text{expected readings for the selected interval}$.
- Warning logic: abnormal-use warnings are produced from transparent rules before any machine-learning extension is considered.

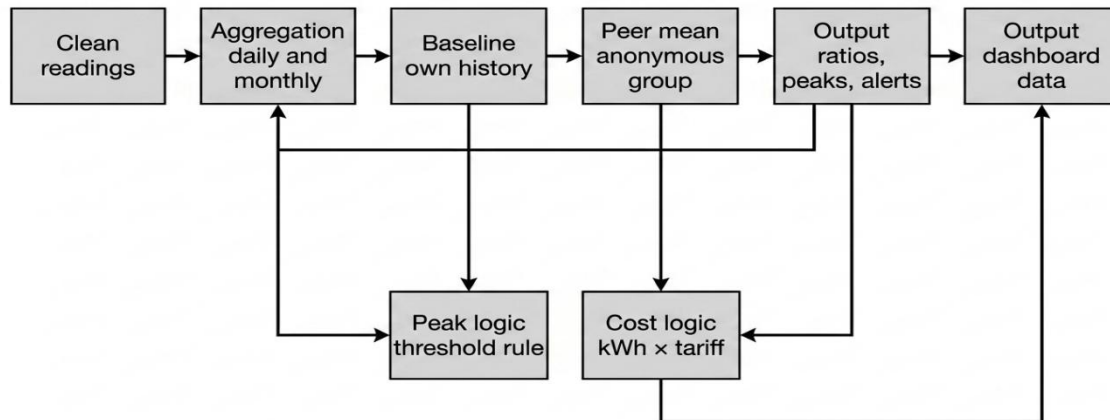


Figure 6:Comparative analytics pipeline.

5. Dashboard interface

The interface will be designed around the decisions a user can make. The landing view will show current or latest consumption, today's kWh, estimated cost, peak power and comparison status. The trend view will show daily and monthly patterns. The device view will rank monitored devices, loads or smart-controller channels by energy consumption. The community view will compare a household or site against an anonymised group average. The alert view will highlight unusual consumption, missing data and configured threshold events.

The dashboard will avoid technical overload. Voltage, current and power are important for the engineering layer, but ordinary users are more likely to respond to kWh, cost, peak period and device ranking. The interface will therefore keep technical variables available while giving priority to practical interpretation.

The dashboard will also show data-quality status. A comparison should not appear reliable if too many readings are missing. The dashboard must therefore show when data is complete, partly missing, duplicated, out of range or simulated as an abnormal event. This prevents the interface from giving a clean-looking answer from poor smart-controller-style data.

Dashboard wireframe for prototype evaluation

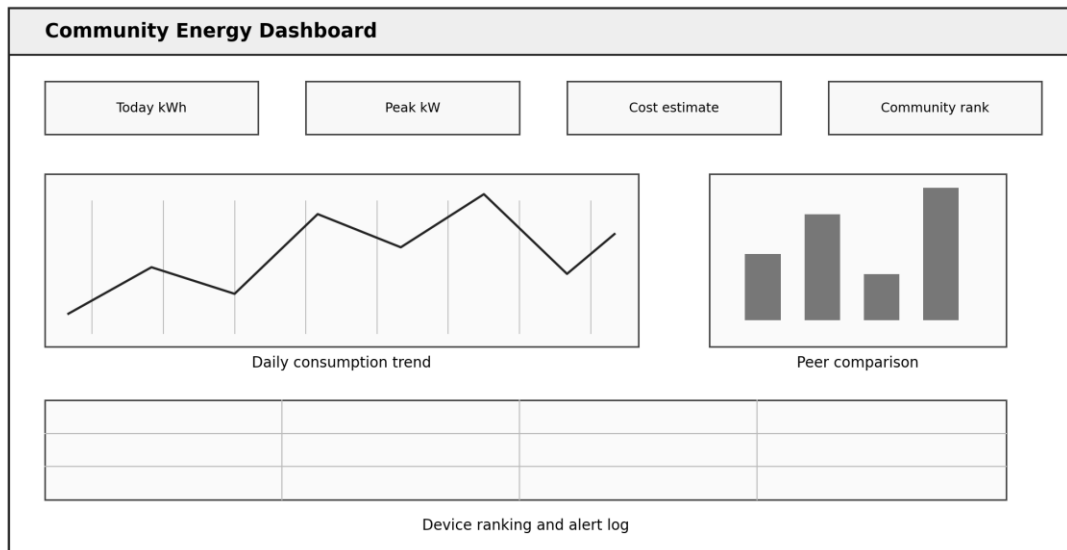


Figure 7:Dashboard wireframe for prototype evaluation.

The dashboard will avoid technical overload. Voltage, current and power are important for the engineering layer, but ordinary users are more likely to respond to kWh, cost, peak period and device ranking. The interface will therefore allow the technical variables to remain available while giving priority to practical interpretation.

6. Implementation tools

The prototype will use open-source tools where possible. A Python backend using FastAPI, Flask or Django can handle ingestion, validation and analytics. SQLite can support early development, while PostgreSQL can be used if the dataset or multi-user testing becomes larger. React, Streamlit or a Python-based dashboard framework can be used for the web interface, depending on development time and integration complexity. Chart libraries will be used for time-series trends, bar comparisons and summary cards. Git will be used for version control.

The tool choice is driven by feasibility rather than novelty. The engineering value lies in the working data pipeline, analytics logic and dashboard evaluation. A simple, stable implementation is preferable to a complex stack that cannot be validated within the project period. Libraries such as scikit-learn and SHAP may be added only if the optional forecasting and explanation extension is implemented after the core dashboard is working.

7. Testing and evaluation

Evaluation will be done at three levels: data correctness, system performance and user interpretation. Data correctness will compare dashboard totals against known simulated smart-controller input totals and manual calculations. System performance will check loading time, update consistency and missing-data handling. User interpretation will check whether users can identify the highest consumption period, the highest consumption device or controller channel and their position relative to the baseline or group average.

The comparison condition should be simple and defensible. Users can first be given raw readings or a spreadsheet-style table, then the dashboard view. The evaluation can then

check whether the dashboard improves interpretation of peak period, high-consumption device, total kWh, estimated cost and peer comparison status. This aligns directly with the hypothesis.

If the optional forecasting or explainable-warning extension is implemented, it will be evaluated separately as a stretch component and not used to judge the success of the core proposal. The project succeeds when the dashboard improves interpretation of structured smart-controller-style energy data through reliable comparative analytics.

Table 6: Testing and evaluation matrix.

Evaluation area	Test method	Success indicator
Data accuracy	Compare dashboard kWh totals with known simulated smart-controller totals and manual calculations.	Error remains within an acceptable tolerance defined before testing.
Data completeness	Calculate valid readings against expected readings for each test period.	Missing intervals are flagged and do not silently distort results.
Analytics correctness	Run known input datasets through baseline, peak and comparison calculations.	Outputs match manual calculations.
Dashboard usability	Small user walkthrough using defined interpretation tasks from raw table view to dashboard view.	Users can interpret key metrics without external explanation.
Responsiveness	Test dashboard on laptop and mobile browser with normal and larger datasets.	Main views load and scale without broken layout.
Privacy-aware presentation	Check identifiers, access roles and aggregate views.	No direct household identity is exposed in community comparisons.
Optional extension evaluation	If forecasting or SHAP-based explanations are implemented, compare warning outputs with known patterns and user understanding.	The extension is useful only if it improves interpretation without weakening the core dashboard.

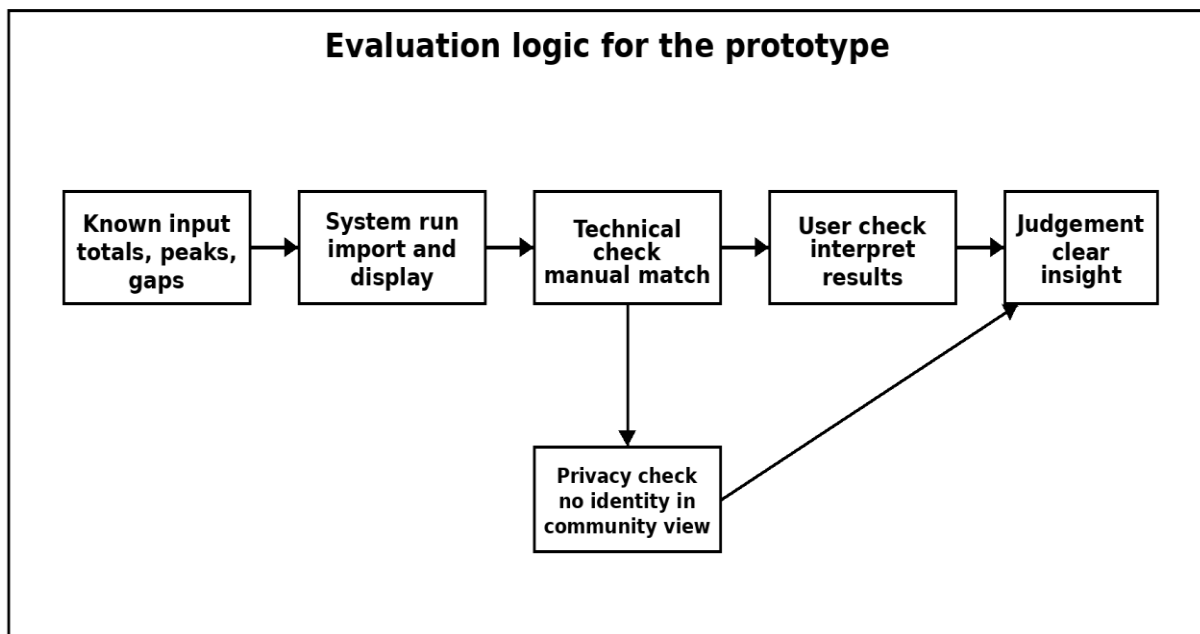


Figure 8:Evaluation logic for the prototype.

8. Ethical and regulatory considerations

The first prototype reduces ethical risk because it uses simulated data. No real household identity, address, or behavioural routine is required for the initial dashboard build. This supports feasibility and avoids unnecessary exposure while the system is still being developed.

The architecture must still be ready for real data because future testing may use approved exports or participant readings. If real household data is introduced, participation will require informed consent. Participants must know what data is collected, why it is collected, how long it will be stored and who can view it. Raw exports must not be shared publicly. Community comparison must remain aggregated or pseudonymised. The project will follow POPIA principles and the CPUT research ethics process where human participation or household data is involved.

The dashboard will not perform control actions on electrical loads during the research phase. The primary scope is monitoring and analytics. Any future load-control feature would need additional safety assessment because switching geysers, pumps or high-current loads can affect user comfort, equipment operation and electrical safety.

9. Risks and mitigation

Table 7: Risks and mitigation.

<i>Risk</i>	<i>Effect on project</i>	<i>Mitigation</i>
<i>Simulated smart-controller data may be seen as less realistic than live readings.</i>	The project may be criticised for not proving live-device integration.	Make the scope clear: the contribution is dashboard analytics and validation. Use realistic profiles, abnormal events and API-ready architecture.
<i>Dataset generation is too simple.</i>	The dashboard may not be tested against meaningful behaviour.	Use site profiles, controller channels, weekday/weekend schedules, variation, missing data and abnormal-use cases.
<i>Dashboard becomes too complex.</i>	Users may not understand the output.	Prioritise kWh, cost, peak period, device ranking and comparison before advanced metrics.
<i>Calculation errors occur in aggregation.</i>	The dashboard may give misleading totals or peaks.	Use manual test cases and known input datasets to verify all formulas.
<i>Privacy concerns arise in future real-data testing.</i>	Data collection may be restricted.	Use consent, pseudonymisation, aggregation, minimal data collection and role-based access.
<i>Tariff differences between municipalities affect cost estimates.</i>	Cost output may be misunderstood.	Use configurable tariffs and label all cost values as estimates.
<i>Hardware or API access becomes available late.</i>	Integration may distract from core objectives.	Keep live integration as a future extension unless it can be added without weakening the validated simulated smart-controller prototype.

10. Expected deliverables

A documented simulated smart-controller data specification.

A controlled simulated smart circuit controller dataset containing timestamped site and device-level readings.

A working data ingestion and storage layer for structured energy readings.

Comparative analytics functions for baseline, group comparison, peak detection, device ranking, data completeness and estimated cost.

A web-based dashboard prototype with individual and community-level views.

A technical evaluation report covering calculation accuracy, usability, responsiveness, limitations and privacy-aware presentation.

A final research report and recommendations for future real-data, export or API integration.

Optional add-on recommendation: a future explainable forecasting module may be specified if sufficient historical data is available, but it will remain outside the minimum deliverable set.

11. Scope and limitations

The scope is limited to monitoring, comparative analytics and visualization using simulated smart circuit controller data. The work does not replace certified electricity meters, official billing, municipal tariff systems or Eskom smart meter infrastructure. The dashboard will estimate consumption and cost from available readings. It will not certify revenue-grade measurements.

The simulated smart-controller dataset can test calculation correctness, interface usability and pipeline reliability, but it cannot prove real household behaviour change. It also cannot prove live-device communication unless a real source is added later. These limitations are acceptable because the corrected project contribution is the dashboard interpretation layer, not commercial smart-device integration.

The community comparison will depend on the number and quality of site profiles in the simulated dataset. Early results will therefore be interpreted as prototype-level evidence rather than population-wide proof. Any forecasting, SHAP-based explanation or recommendation logic will be treated as an add-on or future extension unless the available dataset is strong enough to justify prototype testing within the project period.

12. Work plan

Table 8: Project work plan from February to October 2026.

Criteria / Time frame	Feb	March	April	May	June	July	Aug	Sep	Oct
Topic refinement and proposal approval	X	X							
Literature review and source consolidation	X	X	X						
CBi Astute data source investigation		X	X						
System requirements and architecture design			X	X					

Database ingestion and layer development				X	X				
Analytics algorithm development					X	X			
Dashboard interface development					X	X	X		
Testing with sample and device data						X	X		
Usability and privacy evaluation							X	X	
Final report writing and corrections							X	X	X
Final submission preparation								X	X

13. Conclusion

The project remains feasible because it no longer depends on a specific commercial device or inaccessible API. Its value sits in the engineering chain that can be built and tested directly: simulated smart circuit controller data generation, ingestion, validation, storage, comparative analytics, dashboard visualisation and evaluation. A controlled dataset allows the system to be tested against known totals, known peaks, known missing-data cases, known switching states and known abnormal-use scenarios.

The strongest academic support comes from energy feedback literature, which shows that users need clear and repeated feedback, and from comparative analytics literature, which shows that historical and peer comparison can influence interpretation when handled responsibly. The strongest engineering requirement is to build a reliable data pipeline that can remain source-agnostic. The strongest regulatory requirement is privacy-by-design because real household energy data can reveal behavioural patterns.

A successful prototype will show how structured smart-controller-style energy readings, initially simulated, can be turned into a practical community energy monitoring dashboard that is technically valid, understandable and ethically defensible. Real-device integration can be added later if approved data access becomes available, but it is not required for the core dashboard research to be valid

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